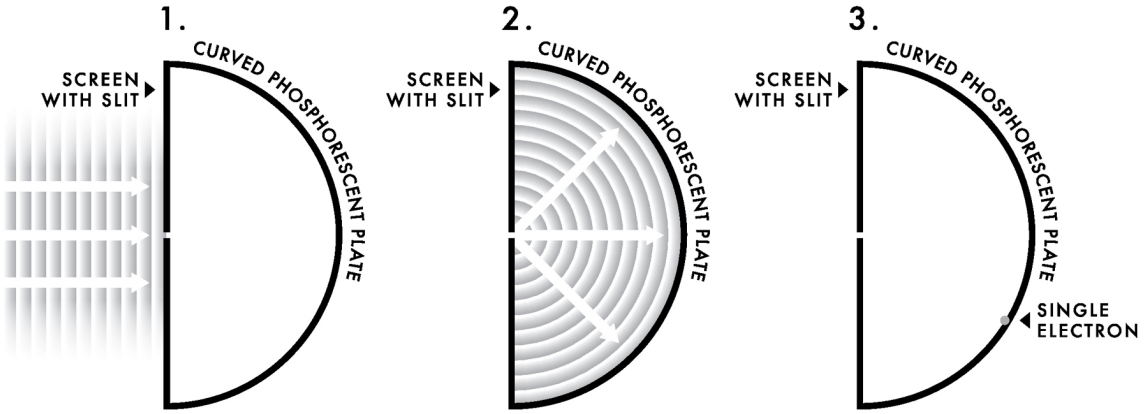


Figure 3.2. Einstein's thought experiment at Solvay. When the electron strikes the plate, how does the rest of the wave function "know" to collapse immediately? After the diagram in Bacciagaluppi and Valentini 2009, p. 486.

Now, Einstein asked his audience to consider the case in which a *single* electron is sent through the hole. Quantum physics still predicts that the electron is equally likely to hit at any location on the screen and cannot be more precise than that. That's fine, though—maybe it just means the theory is incomplete or limited in some way. But, Einstein reminded the group, Heisenberg and Born had claimed that quantum physics was closed, complete, perfect as it was. In that case, there cannot be anything that determines the particular location at which the electron hits the film. But this is a problem—and not because it introduces randomness into nature.

Instead, the problem is one of *locality:* the principle that something that happens in one location can't instantly influence an event that happens somewhere else. The wave function of our single electron is spread evenly across the hemisphere of film, and according to Heisenberg, Born, and Bohr, the electron itself isn't anywhere. The fact that the electron's wave function is evenly spread simply means that the film is equally likely to register the impact of an electron at any location. But, Einstein pointed out, what happens to the wave function when the film does register that impact in one particular spot? Born had shown that a particle's wave function was proportional to the probability of finding the particle in a particular place. But once the electron impacts the film in a particular spot, the probability of it hitting the film anywhere else immediately drops to zero. So, somehow,